Department of Computer and Information Systems Engineering

CS-406 Computer Engineering Project

Proposal for the Final Year Design Project

	Title	**SEE BEYOND – Guided by sound empowered by touch** “An assistive smart glasses and cane system that combines real-time vision AI with audio and haptic feedback to empower visually impaired individuals with safer navigation and independent living”

Domain	Domain 1 Computer Vision	Domain 2 Machine Learning	Domain 3 Mobile Development	Domain 4 Database Management	Domain 5 Digital System Design	Domain 6

1. Nature of Project [Tick all that applicable]

☒ New Project OR ☐ Extension of Existing Project	☐ Industrial Collaboration	☐ Funded
☐ Other Department Collaboration (If yes) Department Name_________________	☐ Other Academic Institution Collaboration (If yes) Institution Name________________	

2. Brief Outline (*Problem Identification and Significance*)

Navigating the world without sight presents profound challenges for individuals with visual impairments. Everyday activities such as crossing busy streets, handling money, reading important signs, or detecting unexpected obstacles can quickly become overwhelming, unsafe, or even impossible without assistance. These challenges often lead to reduced independence, limited mobility, social isolation, and a loss of confidence in one’s ability to engage with the environment freely and safely.

Although tools like white canes, guide dogs, and mobile applications have long provided support, they typically serve only one function at a time and often fall short in delivering full situational awareness.

To address this gap, **See Beyond--**a smart, AI-powered solution brings an innovative solution. The system aims to integrate essential capabilities such as obstacle detection, text and currency recognition, people recognition, scenes and object detection, and instant voice command into a single, intuitive platform. By processing data locally on the device offline, it offers practical assistance in various real-life scenarios, empowering visually impaired users to navigate the world with confidence and ease.

3. Objectives

The primary objectives of the **See Beyond** project are:

1. **Improve Situational Awareness**
   To provide real-time awareness of obstacles, people, text, and objects in the surroundings, thereby reducing safety risks and fostering user confidence in daily activities.

2. **Facilitate Seamless Human–Device Interaction**
   To design an intuitive interaction model through natural voice commands and audio feedback, ensuring accessibility even for users with limited technical literacy.
3. **Promote Financial and Social Inclusion**
   To support essential daily tasks such as identifying currency, reading text, and recognizing acquaintances, thereby reducing reliance on others and increasing participation in social and economic activities.
4. **Ensure Accessibility Through Affordable Technology**
   To engineer a compact, low-cost, and energy-efficient assistive device that is deployable in real-world conditions, making advanced technology accessible to marginalized communities.
5. **Enable Secure and Scalable Data Management**
   To provide mechanisms for authorized relatives or caregivers to manage recognition databases securely, ensuring adaptability and long-term usability of the system.

4. Scope

The scope of **See Beyond** encompasses the design and development of an AI-powered assistive system tailored for visually impaired individuals, with the overarching aim of enhancing mobility, safety, and independence. The project focuses on creating a lightweight computer vision–based module capable of real-time object and obstacle detection to ensure reliable navigation in complex environments. It further includes the integration of multilingual text recognition and Pakistani currency identification using OCR and computer vision algorithms, along with a secure, facial recognition system for identifying pre-registered acquaintances through a companion mobile application. To facilitate natural and accessible human–device interaction, the system will incorporate a voice-enabled interface with audio feedback delivered via Wi-Fi. The project also involves the engineering of a compact, battery-efficient embedded hardware prototype optimized for offline performance, as well as the development of a mobile application that allows authorized family members or caregivers to update the recognition database remotely and securely.

5. Proposed Methodology

The project will follow an iterative development methodology combining hardware-software co-design, user-centered design, and software development practices. The key stages are:

1. **Requirement Analysis and Literature Review**
   - Study the existing assistive technologies for the visually impaired.
   - Gather input from target users and experts to determine essential features and pain points.
   - Define technical and usability requirements.

2. **Hardware Design and Selection**

- Choose suitable **microcontroller (ESP32).**
- Integrate peripherals like camera module, battery and audio output.

3. **Software Development**
   - **Obstacle & Object Detection:** Train and deploy lightweight object detection models (e.g., YOLOv5 Nano or MobileNet SSD) on embedded hardware.
   - **People Recognition:** Implement a local facial recognition system using image comparison techniques (FaceNet) with a custom offline database.
   - **OCR & Currency Detection:** Use Tesseract OCR for multilingual text extraction and a trained CNN model for Pakistani currency classification.
   - **Voice Interface:** Integrate text-to-speech (for audio feedback) modules.

4. **Mobile Application Development**
   - Build a user-friendly mobile app using Flutter.
   - Enable secure login and allow family members to update known faces and names.
   - Sync database changes with the device via local or cloud connectivity.

5. **System Integration and Testing**
   - Integrate hardware and software into a working prototype.
   - Conduct unit testing, integration testing, and field testing with real users.
   - Gather user feedback and refine the system.

6. **Optimization and Finalization**
   - Optimize energy consumption and processing performance.
   - Improve UX based on feedback and usability studies.
   - Prototype Development and Integration
   - Assemble the hardware components into a wearable, portable device.
   - Integrate software modules for seamless communication and processing.
   - Finalize prototype and prepare documentation for deployment or future scale-up**.**

7. **Testing and Validation**
   - Perform component-level and system-level testing.
   - Conduct real-life testing with target users to evaluate performance, usability, and reliability.
   - Gather feedback for system improvement.

8. **Deployment and Documentation**
   - Finalize the prototype and app.
   - Document the development process, system usage, and setup instructions for users**.**

Project Deliverables:

The proposed system, **See Beyond,** will deliver the following technological outcomes:

1. **AI-powered Real-time Object and Obstacle Detection**
   Development of a computer vision–based module capable of identifying and classifying physical objects and obstacles in the user's environment, ensuring safe and efficient navigation. This obstacle detection functionality is implemented using the microcontroller **ESP32**, **VL53L1X**

sensor with real-time distance measurement is used to detect presence and proximity of obstacles within a defined range.

2. **Voice-enabled Personal Assistant Interface**
   Implementation of a natural language interaction module, similar to Alexa, enabling users to issue spoken commands (e.g., requesting directions from one location to another or obtaining contextual descriptions of surroundings).
3. **OCR-based Text-to-Speech Translation**
   Integration of Optical Character Recognition (OCR) with text-to-speech conversion, allowing users to access printed material, signage, and documents through audio output.
4. **Currency Recognition System**
   Deployment of a convolutional neural network (CNN) trained for Pakistani banknotes, enabling accurate recognition of currency denominations and promoting financial independence for visually impaired users.
5. **People Recognition**
   Implementation of a secure, locally hosted facial recognition database to identify pre-registered acquaintances, ensuring privacy and reliable offline operation, using **tensor flow**.
6. **Mobile Application with Cloud Synchronization**
   Design of a user-friendly mobile application with secure login, cloud synchronization, and database management features, allowing authorized family members to update the recognition system remotely.
7. **Compact, Low-power Embedded System Prototype**
   Engineering of a wearable embedded hardware prototype that balances computational performance and energy efficiency, incorporating Bluetooth or Wi-Fi connectivity for seamless interaction with mobile devices.

6. Resources Involved

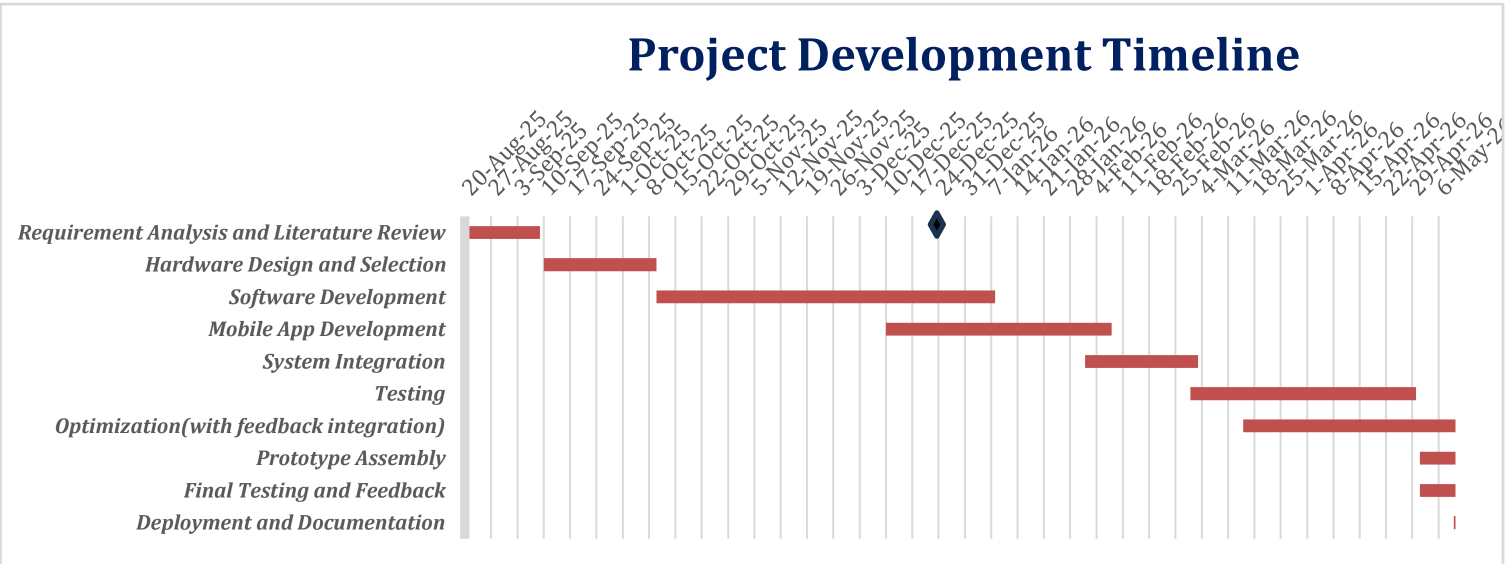
1. **Hardware Resources**
   - Edge AI processing platform (ESP32).
   - Camera modules for image and video capture.
   - VL53L1X sensors for obstacle detection.
   - Microphones and speakers/earphones for audio interaction.
   - Power supply components (battery packs, charging modules).
   - Wireless communication modules (Bluetooth/Wi-Fi).
2. **Software Resources**
   - Python, TensorFlow, PyTorch (for AI model development).
   - OpenCV (for computer vision tasks).
   - Tesseract OCR (for text recognition).
   - Flutter (for mobile application development).
   - Database tools (SQL) for secure storage and synchronization.
3. **Testing and Development Resources**
   - Development environment (laptops/PCs with GPU support(optional) for training).
   - Prototyping kits and soldering tools for hardware integration.
   - Field-testing environments (indoor and outdoor navigation scenarios).

7. Description of Industrial Support (If any)

8. SDGs (If Applicable)

☐ No Poverty	☐ Zero Hunger
☒ Good Health and Well-Being	☐Quality Education
☐ Gender Equality	☐ Clean water and Sanitation
☐ Affordable and Clean Energy	☐ Decent Work and Economic growth
☒ Industry, Innovations and Infrastructure	☒ Reduced Inequalities
☐ Sustainable Cities and Communities	☐ Responsible Consumption and Production
☐ Climate action	☐ Life Below Water
☐ Life on Land	☐ Peace, Justice and Strong Institutions
☐ Partnerships	

9. Gantt Chart



◆ **MILESTONE:**

Our MVP will consist of a **smart cane** equipped with obstacle detection functionality, offering three selectable detection ranges to support indoor and outdoor navigation. In addition, a **mobile application** will be developed to support key assistive features, including **currency detection, object recognition, and people recognition**.

10. Details of Project Team

i. Students

No.	Name	Seat No.	Signature (s)
1	Sheikh Malaika Mustafa	CS-22006	
2	Sumaika Imran	CS-22009	
3	QurratulAin	CS-22018	
4	Urooj Fatima	CS-22022	

ii. Supervisors / Advisors

	Name	Designation & Department	Address & Contact	Signature(
Supervisor	Dr.Urooj Ainuddin	Assistant Professor ,CIS		
Co-Supervisor (If any)	-	-		
Industrial Advisor (If any)				

For Office Use Only		
Project Serial No.: ___________ Dated: ________________	Signature Convener Steering Committee	Signature FYP Coordinator

☐ Proposal Approved	☐ Not Approved	☐ Returned for Clarification / Modification
Comments: (if any)		

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(Signature of Chairperson)

Date: ______________________________